Physics Part 6

Mirrors & Lenses

Continuation…

SIGN CONVENTION FOR SPHERICAL MIRRORS

Quantity	When Positive	When Negative
Object distance, o	Always	Never
Image distance, i	Real image	Virtual image
Focal length, f	Converging mirror (concave) $f = \frac{1}{2}r$	Diverging mirror (convex) $f = -\frac{1}{2}r$
Magnification, m	Upright image	Inverted image

Properties of Concave Mirrors (Converging):

1. As the object is moved closer, image distance increase and image height increases.
2. When the object is at the focal length, there is no image.
3. Within the focal length, the image is upright, virtual.
4. As the object approaches the mirror, image height becomes very close to the object height.

Properties of Convex Mirrors (Diverging):

1. Field of view of convex mirrors is wider
2. Image is always virtual, upright and smaller
3. As object approaches the mirror, the virtual image on the other side approaches the mirror and becomes larger

Thin Lenses

When a ray of light passes through a lens, it bends toward the thicker part of the lens if the lens has an index of refraction greater than that of the surrounding medium. **Principal focus** is the point through which rays parallel to the principal axis of the lens pass. **Focal length** is the distance from the lens to the principal focus. **Optical center** is a point on the principal axis through which rays pass without changing direction. When the rays actually pass through the image after refraction, the image formed is known as **real image**. When the rays only appear to come from the image after refraction, the image is known as **virtual image**.

Converging lens makes a set of parallel rays converge after refraction. It forms real images when the object is farther from the lens than the principal focus and virtual images when the object is between the lens and the principal focus. **Diverging lens** makes a set of parallel rays divergent after refraction. It forms a virtual image of any object. The position, size and nature of an image can be determined by the use of the following lens equation:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

where:
p - object distance q - image distance
f - focal length

For the sign convention for lens, it is exactly the same as the sign convention for mirrors.

Linear magnification is the ratio of the size of the image to the size of the object.

$$M = \frac{\text{image size}}{\text{object size}} = -\frac{q}{p}$$

Power of a lens (D) is the amount by which the lens can change the curvature of a wave. It is equivalent to the reciprocal of the lens' focal length. The unit of the power of a lens is **diopter**.

$$P = \frac{1}{f}$$

Properties of Concave Lenses (Diverging):

1. Image produced by diverging lenses are always upright, virtual (same side of lens as object) and reduced in size.
2. As the object moves closer to the lens, the image moves closer to the lens as well.
3. As the object moves closer to the lens, the image size is increased.

Properties of Convex Lenses (Converging):

1. As object is moved closer to the lens, the image distance and height increases.
2. At twice the focal length, image and object distance and height are equal.
3. As object approaches one focal length, image distance and height approaches infinity
4. When object is at focal length, there is no image.
5. Within the focal length, image is upright, virtual and same side as object.

Lensmaker's Equation

The lensmaker's equation is used to relate the radii of curvature, the thickness, the refractive index and the focal length of a thick lens.

For thick lens: $$\frac{1}{f} = (n-1)\left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n-1)d}{nR_1R_2}\right]$$

where:
f = focal length
R_1 and R_2 = radii of curvature
n = index of refraction
d = thickness of lens

For thin lens: $$\frac{1}{f} = (n-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

The focal length f is positive for converging lenses, and negative for diverging lenses. The reciprocal of the focal length, 1/f, is the optical of the lens. If the focal length is in meters, this gives the optical power in diopters (inverse meters).

The signs of R_1 and R_2 indicate whether the corresponding surfaces are convex or concave. If R1 is positive the first surface is convex, and if R1 is negative the surface is concave. The signs are reversed for the back surface of the lens: if R2 is positive the surface is concave, and if R2 is negative the surface is convex. If either radius is infinite, the corresponding surface is flat.

Key words

Aberration the convergence to different foci, by a lens or mirror, of rays of light emanating from one and the same point, or the deviation of such rays from a single focus; a defect in a focusing mechanism that prevents the intended focal point

Achromatic doublet a type of lens made up of two simple lenses paired together designed so that the chromatic aberration of each lens partially offsets the other; in this way light in a range of wavelengths may be brought to the same focus

Afocal system an optical system that produces no net convergence or divergence of the beam, i.e. has an infinite effective focal length

Thin lens one whose thickness allows rays to refract but does not allow properties such as dispersion and aberrations.

Airy pattern the diffraction pattern formed by a perfect lens with a circular aperture, imaging a point source

Coma an off-axis lens aberration resulting from a variation of lens focal lengths as a function of aperture zone or annulus

Complex lens a lens assembly consisting of several compound lenses

Compound lens a lens assembly consisting of a number of simple lens elements

Condenser lens a lens assembly designed to collect energy from a light source

Diffraction limited lens a lens with negligible residual aberrations

Dispersion the variation in the refractive index of a medium as a function of wavelength; the property of an optical system which causes the separation of mononochromatic components of radiation.

Effective focal length the distance from the principal point to the focal point

Hyperfocal the distance at which a lens may be focused to produce satisfactory image quality over an extended range of object distances.

Image distance the distance of the image from the center of the lens

Petzval curvature the natural tendency for a lens to produce its final image on a curved rather than flat surface

Ray tracing a technique used in optics for analysis of optical systems

Surface vertices the points where each surface crosses the optical axis

Thick lens lenses whose thickness are not negligible, i.e. one cannot make a simple assumption that a light ray is refracted only once in the lens

Thin lens one whose thickness allows rays to refract but does not allow properties such as dispersion and aberrations.

Throughput a measure of the efficiency of an optical system in terms of the luminous flux collected and delivered